Testing

1. Change History

Change Date	Modified Sections	Rationale
2026-10-17	All	First version for CP1: interface tests for the Auth & Profile Service, Activity & Review Service and Itinerary Engine, the NFR 1 test, and the Generate Itinerary end-to-end test design.
2026-10-26	2.1.1	Added the Group Service interfaces and group trips in the Itinerary Engine interfaces.
2026-10-26	2.1.1, 3	Added the adjust interface, Socket.io live updates, and the NFR 2 test with its log.
2026-10-26	2.1.1	Added trip options to the generate tests (Google Routes mocked and through the production client) and the `add` adjustment.
2026-10-26	2.1.1	Added the Friend Service interfaces.
2026-10-26	2.1.1	Added the Feed Service interfaces; copying is also tested with Google Routes mocked.
2026-10-26	2.1.1	Added the `visit` and `unvisit` adjustments and Socket.io location sharing.
2026-10-26	2.1.1	Added the Nearby Service interfaces.
2026-10-26	2.1.1	Added verified reviews and the reviews list interface.
2026-10-26	2.1.1	Added the suggestion interfaces, the disruption timer (OpenWeather and Places mocked and through the production clients) and the `suggestionAdded` push.
2026-10-26	2.1.1, 2.3-2.5, 3, 4	Added three unmocked tests and regenerated coverage, with uncovered lines now mapped to the TypeScript source. Added the NFR 3 test, and implemented the three end-to-end tests with their logs.
2026-10-27	4.2	The Generate Itinerary test's unknown lodging is now "no such place zzzz", because Google Places matched the old one to real Vancouver hotels.
2026-11-02	All	Shortened the wording after CP1 feedback, and renumbered the Generate Itinerary test steps to match the CP2 use case.

2. Back-end Test Specification: APIs

2.1. Locations of Back-end Tests and Instructions to Run Them

2.1.1. Tests

Interface	Describe Group Location, No Mocks	Describe Group Location, With Mocks	Mocked Components
POST /api/auth/google	`backend/tests/no-mock/auth.test.ts#L13`	`backend/tests/mock/auth.test.ts#L29`	Google Identity Services (token verification), User DB

Interface	Describe Group Location, No Mocks	Describe Group Location, With Mocks	Mocked Components
GET /api/users/me	backend/tests/no-mock/auth.test.ts#L60	backend/tests/mock/auth.test.ts#L173	User DB
PUT /api/users/me	backend/tests/no-mock/auth.test.ts#L153	backend/tests/mock/auth.test.ts#L195	User DB
GET /api/activities	backend/tests/no-mock/activities.test.ts#L22	backend/tests/mock/activities.test.ts#L46, #L201 (through the production Google Places client)	Google Places API, Activity DB
POST /api/activities/:activityId/reviews	backend/tests/no-mock/reviews.test.ts#L56	backend/tests/mock/reviews.test.ts#L27	Review DB, Activity DB, City catalogue DB
GET /api/activities/:activityId/reviews	backend/tests/no-mock/reviews.test.ts#L314	backend/tests/mock/reviews.test.ts#L112	Review DB
POST /api/groups	backend/tests/no-mock/groups.test.ts#L23	backend/tests/mock/groups.test.ts#L25	Group DB (invite-code collisions, write failures)
GET /api/groups	backend/tests/no-mock/groups.test.ts#L73	backend/tests/mock/groups.test.ts#L78	Group DB
POST /api/groups/join	backend/tests/no-mock/groups.test.ts#L116	backend/tests/mock/groups.test.ts#L97	Group DB
GET /api/groups/:groupId	backend/tests/no-mock/groups.test.ts#L207	backend/tests/mock/groups.test.ts#L118	Group DB
PUT /api/groups/:groupId/members/me	backend/tests/no-mock/groups.test.ts#L245	backend/tests/mock/groups.test.ts#L137	Group DB (member removed during the update)
DELETE /api/groups/:groupId/members/:userId	backend/tests/no-mock/groups.test.ts#L304	backend/tests/mock/groups.test.ts#L157	Group DB
DELETE /api/groups/:groupId	backend/tests/no-mock/groups.test.ts#L377	backend/tests/mock/groups.test.ts#L179	Itinerary DB
POST /api/itineraries/generate	backend/tests/no-mock/itineraries.test.ts#L38	backend/tests/mock/itineraries.test.ts#L37, #L198 (Google Routes travel times, cache and failures), #L284 (through the production Google Routes client)	Google Places API (lodging lookup), Google Routes API, Itinerary DB
GET /api/itineraries/:itineraryId	backend/tests/no-mock/itineraries.test.ts#L393	backend/tests/mock/itineraries.test.ts#L136	Itinerary DB
GET /api/itineraries	backend/tests/no-mock/itineraries.test.ts#L442	backend/tests/mock/itineraries.test.ts#L157	Itinerary DB

Interface	Describe Group Location, No Mocks	Describe Group Location, With Mocks	Mocked Components
POST /api/itineraries/:itineraryId/adjust	backend/tests/no-mock/adjust.test.ts#L41	backend/tests/mock/adjust.test.ts#L32	Itinerary DB (a concurrent change saved first, write failure), Activity DB, live-update push
Socket.io itinerary updates	backend/tests/no-mock/tripSync.test.ts#L40	backend/tests/mock/tripSync.test.ts#L19	Itinerary DB (access check on subscribe)
Socket.io location sharing	backend/tests/no-mock/tripSync.test.ts#L144	backend/tests/mock/tripSync.test.ts#L46	User DB (looking up the sharer)
Socket.io suggestions	backend/tests/no-mock/tripSync.test.ts#L229	backend/tests/mock/tripSync.test.ts#L75	Suggestion push (a device that cannot be reached)
GET /api/travellers	backend/tests/no-mock/friends.test.ts#L37	backend/tests/mock/friends.test.ts#L33	User DB
GET /api/travellers/:userId	backend/tests/no-mock/friends.test.ts#L133	backend/tests/mock/friends.test.ts#L51	Friendship DB
GET /api/friends	backend/tests/no-mock/friends.test.ts#L186	backend/tests/mock/friends.test.ts#L69	Friendship DB
POST /api/friends/requests	backend/tests/no-mock/friends.test.ts#L236	backend/tests/mock/friends.test.ts#L87	Friendship DB (both travellers asking at the same moment, write failure)
POST /api/friends/requests/:userId/accept	backend/tests/no-mock/friends.test.ts#L292	backend/tests/mock/friends.test.ts#L126	Friendship DB
DELETE /api/friends/:userId	backend/tests/no-mock/friends.test.ts#L327	backend/tests/mock/friends.test.ts#L145	Friendship DB
PUT /api/friends/privacy	backend/tests/no-mock/friends.test.ts#L361	backend/tests/mock/friends.test.ts#L164	User DB
POST /api/posts	backend/tests/no-mock/posts.test.ts#L55	backend/tests/mock/posts.test.ts#L54	Post DB
GET /api/posts	backend/tests/no-mock/posts.test.ts#L144	backend/tests/mock/posts.test.ts#L72	Comment DB
GET /api/posts/:postId	backend/tests/no-mock/posts.test.ts#L238	backend/tests/mock/posts.test.ts#L90	Comment DB
DELETE /api/posts/:postId	backend/tests/no-mock/posts.test.ts#L275	backend/tests/mock/posts.test.ts#L107	Comment DB
PUT /api/posts/:postId/like	backend/tests/no-mock/posts.test.ts#L311	backend/tests/mock/posts.test.ts#L125	Post DB (post deleted while liking it)

Interface	Describe Group Location, No Mocks	Describe Group Location, With Mocks	Mocked Components
DELETE /api/posts/:postId/like	backend/tests/no-mock/posts.test.ts#L345	backend/tests/mock/posts.test.ts#L143	Post DB
GET /api/posts/:postId/comments	backend/tests/no-mock/posts.test.ts#L365	backend/tests/mock/posts.test.ts#L160	Comment DB
POST /api/posts/:postId/comments	backend/tests/no-mock/posts.test.ts#L403	backend/tests/mock/posts.test.ts#L177	Comment DB
DELETE /api/posts/:postId/comments/:commentId	backend/tests/no-mock/posts.test.ts#L434	backend/tests/mock/posts.test.ts#L194	Comment DB
POST /api/posts/:postId/copy	backend/tests/no-mock/posts.test.ts#L472	backend/tests/mock/posts.test.ts#L213	Google Routes API, Itinerary DB
PUT /api/nearby/trips/:itineraryId	backend/tests/no-mock/nearby.test.ts#L52	backend/tests/mock/nearby.test.ts#L75	Itinerary DB
GET /api/nearby/trips	backend/tests/no-mock/nearby.test.ts#L110	backend/tests/mock/nearby.test.ts#L41	Itinerary DB, Group DB (a group deleted while listing)
POST /api/nearby/trips/:itineraryId/requests	backend/tests/no-mock/nearby.test.ts#L194	backend/tests/mock/nearby.test.ts#L92	Join request DB
GET /api/nearby/requests	backend/tests/no-mock/nearby.test.ts#L247	backend/tests/mock/nearby.test.ts#L109	Join request DB
POST /api/nearby/requests/:requestId/accept	backend/tests/no-mock/nearby.test.ts#L287	backend/tests/mock/nearby.test.ts#L126	Group DB (the group fills up first)
POST /api/nearby/requests/:requestId/decline	backend/tests/no-mock/nearby.test.ts#L345	backend/tests/mock/nearby.test.ts#L146	Join request DB
GET /api/suggestions	backend/tests/no-mock/suggestions.test.ts#L43	backend/tests/mock/suggestions.test.ts#L151	Suggestion DB
POST /api/suggestions/:suggestionId/accept	backend/tests/no-mock/suggestions.test.ts#L106	backend/tests/mock/suggestions.test.ts#L181	Itinerary DB (a change saved at the same moment), Suggestion DB
POST /api/suggestions/:suggestionId/dismiss	backend/tests/no-mock/suggestions.test.ts#L169	backend/tests/mock/suggestions.test.ts#L217	Suggestion DB
Disruption timer: checkActiveTrips()	backend/tests/no-mock/suggestions.test.ts#L191	backend/tests/mock/suggestions.test.ts#L63, #L254 (through the production OpenWeather and Places clients)	OpenWeather API, Google Places API (business status), Suggestion DB
GET /health	backend/tests/no-mock/health.test.ts#L10	backend/tests/mock/health.test.ts#L11	MongoDB connection state

The tests without mocks run against a real in-memory MongoDB (`backend/tests/helpers/testDatabase.ts`) preloaded with the Vancouver catalogue, so they need no Google Places key. The engine's scheduling core also has unit tests in `backend/tests/unit/`, including a

comparison against a brute-forced optimum.

2.1.2. Commit Hash Where Tests Run

`ceec5d99d70ad8d0d26c8d107e0d631a3d966fd9`

2.1.3. Explanation on How to Run the Tests

1. **Clone the repository** and install the back-end dependencies (Node.js 22 and npm 10; Docker is not needed):

   ```
   git clone https://github.com/abhiverma13/TimeTravel.git
   cd TimeTravel/backend
   npm ci
   ```

2. **Run the interface tests** (with mocks, without mocks and combined, each with coverage) from the repository root:

   ```
   ./scripts/run-backend-interface-tests.sh      # or .\scripts\run-backend-interface-tests.ps1 on Windows
   ```

3. **Run the NFR tests**:

   ```
   ./scripts/run-backend-nfr-tests.sh            # or .\scripts\run-backend-nfr-tests.ps1 on Windows
   ```

4. **Run everything** (interface, unit and NFR tests): `cd backend && npm test`.

The first run downloads a MongoDB binary (about 1 minute). Coverage reports go to `backend/coverage/mock`, `backend/coverage/no-mock` and `backend/coverage/interface`.

2.2. GitHub Actions Configuration Location

`.github/workflows/backend-tests.yml`: on every push and pull request to `main`, it type-checks the back-end, runs the interface (with coverage), unit and NFR tests, and uploads the coverage reports.

2.3. Jest Coverage Report Screenshots With Mocks

All files

81.78% Statements `1634/1998` **58.86%** Branches `508/863` **78.8%** Functions `394/500` **83.73%** Lines `1550/1851`

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
src		100%	29/29	50%	1/2	100%	2/2	100%	29/29
src/activities		88.15%	253/287	68.7%	101/147	92.59%	50/54	88.72%	244/275
src/auth		85.33%	64/75	59.09%	13/22	88.23%	15/17	86.48%	64/74
src/config		0%	0/8	0%	0/11	0%	0/2	0%	0/8
src/db		85.71%	6/7	100%	0/0	66.66%	2/3	85.71%	6/7
src/disruptions		94.15%	161/171	88.31%	68/77	94.28%	33/35	94.54%	156/165
src/friends		73.43%	94/128	47.91%	23/48	72.22%	26/36	76.85%	93/121
src/groups		76.22%	109/143	53.33%	24/45	72.5%	29/40	78.78%	104/132
src/itineraries		75.19%	382/508	46.59%	123/264	71.72%	104/145	77.55%	349/450
src/itineraries/engine		94.21%	228/242	81.73%	85/104	96.15%	50/52	94.59%	210/222
src/middleware		69.04%	29/42	47.36%	9/19	75%	6/8	70.73%	29/41
src/nearby		78.41%	109/139	51.78%	29/56	69.04%	29/42	84.16%	101/120
src/posts		85.03%	125/147	52.5%	21/40	86.66%	39/45	87.59%	120/137
src/shared		100%	1/1	100%	0/0	100%	0/0	100%	1/1
src/sync		61.97%	44/71	39.28%	11/28	47.36%	9/19	63.76%	44/69

Code coverage generated by istanbul at 2026-10-26T23:41:48.612Z

```
File                     | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------|---------|----------|---------|---------|------------------
All files                |   81.78 |    58.86 |    78.8 |   83.73 |
  app.ts                 |     100 |       50 |     100 |     100 | 38
  activityModel.ts       |     100 |      100 |     100 |     100 |
  activityRoutes.ts      |     100 |       50 |     100 |     100 | 47
  activityService.ts     |   81.73 |    66.66 |   86.95 |   83.33 | 143,155-164,169-170,196,204,209-211,265
  communityScore.ts      |     100 |       80 |     100 |     100 | 25-26
  interests.ts           |     100 |    66.66 |     100 |     100 | 85-86
  openingHours.ts        |     100 |      100 |     100 |     100 |
  placesClient.ts        |     100 |    76.92 |     100 |     100 | 116-122,141
  reviewModel.ts         |      80 |       50 |     100 |      80 | 76,79,82,85
  reviewTags.ts          |   76.31 |    42.85 |   83.33 |   76.31 | 40-41,60-61,71,75-76,80,85
  authRoutes.ts          |     100 |       50 |     100 |     100 | 15
  authService.ts         |   73.52 |    42.85 |     100 |   73.52 | 50,53,60-63,73,76-79
  googleTokenVerifier.ts |     100 |      100 |     100 |     100 |
  userModel.ts           |      75 |      100 |   33.33 |   85.71 | 44
  env.ts                 |       0 |        0 |       0 |       0 | 1-16
  connection.ts          |   85.71 |      100 |   66.66 |   85.71 | 12
  disruptionRoutes.ts    |   93.75 |      100 |     100 |   93.75 | 30
  disruptionService.ts   |   92.74 |    88.73 |   92.59 |   93.22 | 188,204,226,229,232,235-237
  suggestionModel.ts     |     100 |      100 |     100 |     100 |
  weatherClient.ts       |     100 |    83.33 |     100 |     100 | 74
  friendRoutes.ts        |   97.14 |      100 |     100 |   97.05 | 67
  friendService.ts       |   61.17 |    43.18 |      60 |   65.82 | 61,64,67,90,96,102-108,115,131-132,144-147,153-154,160-
163,183,189,200-202
  friendshipModel.ts     |     100 |      100 |     100 |     100 |
  groupModel.ts          |     100 |      100 |     100 |     100 |
  groupRoutes.ts         |   91.17 |      100 |   88.88 |   91.17 | 21,64,69
  groupService.ts        |    69.6 |    53.33 |   67.74 |   72.52 | 73-84,118,135,138,172,182,190,193,196,205,208-
219,231,235,258
  dayRepair.ts           |   33.57 |    18.64 |    37.2 |   35.65 | 58-61,76-87,119,137-140,154,165-284,299-307
  itineraryModel.ts      |     100 |      100 |     100 |     100 |
  itineraryRoutes.ts     |   86.95 |       50 |   77.77 |   90.69 | 43,45,48,51
  itineraryService.ts    |    87.3 |    64.51 |   81.66 |   87.86 | 145-146,210-216,295,315,333-334,361-
368,423,467,472,505,512,520,588,592
  routesClient.ts        |     100 |    92.85 |     100 |     100 | 70
  travelTimes.ts         |     100 |    85.71 |     100 |     100 | 90,105
  tripCalendar.ts        |     100 |       60 |     100 |     100 | 23
  tripOptions.ts         |     100 |       60 |     100 |     100 | 16
  visits.ts              |   77.77 |      100 |      50 |   77.77 | 11-14
  adjust.ts              |   68.42 |    66.66 |      50 |   72.22 | 16-20,37
  candidates.ts          |   85.71 |    70.83 |   92.85 |   87.09 | 40-44
  daySchedule.ts         |     100 |    92.85 |     100 |     100 | 59
  localSearch.ts         |     100 |    95.23 |     100 |     100 | 18
  planner.ts             |   97.36 |    79.31 |     100 |   97.14 | 16,20
  travelTime.ts          |   95.65 |       80 |     100 |   95.65 | 41
  types.ts               |     100 |      100 |     100 |     100 |
  authenticate.ts        |   76.47 |       50 |     100 |   76.47 | 12-13,19,28
  errorHandler.ts        |      64 |    46.15 |      60 |   66.66 | 20,26-27,34-36,39-40
  joinRequestModel.ts    |     100 |      100 |     100 |     100 |
  nearbyRoutes.ts        |     100 |      100 |     100 |     100 |
  nearbyService.ts       |   70.87 |    51.78 |   61.76 |   77.64 | 93-96,101,104,123,135-146,159,163,167,177,180-
181,189,192,195,203
  postModel.ts           |     100 |      100 |     100 |     100 |
  postRoutes.ts          |   93.47 |       50 |     100 |   93.18 | 56,65,87
  postService.ts         |   78.16 |    52.63 |   81.25 |   82.27 | 107,127,145,155,166,169,180,187-188,202,205,222,232,247
  interestTags.ts        |     100 |      100 |     100 |     100 |
  tripSync.ts            |   61.97 |    39.28 |   47.36 |   63.76 | 63-64,83-84,88-89,98-106,130,136,140-144,150,154-
156,161,166,169
```

2.4. Jest Coverage Report Screenshots Without Mocks

All files

91.14% Statements `1821/1998` **74.5%** Branches `643/863` **92.8%** Functions `464/500` **90.81%** Lines `1681/1851`

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
src		100%	29/29	50%	1/2	100%	2/2	100%	29/29
src/activities		86.75%	249/287	63.26%	93/147	85.18%	46/54	86.54%	238/275
src/auth		90.66%	68/75	72.72%	16/22	100%	17/17	90.54%	67/74
src/config		0%	0/8	0%	0/11	0%	0/2	0%	0/8
src/db		100%	7/7	100%	0/0	100%	3/3	100%	7/7
src/disruptions		73.68%	126/171	51.94%	40/77	80%	28/35	73.33%	121/165
src/friends		97.65%	125/128	87.5%	42/48	97.22%	35/36	97.52%	118/121
src/groups		96.5%	138/143	77.77%	35/45	97.5%	39/40	96.21%	127/132
src/itineraries		89.17%	453/508	72.34%	191/264	91.03%	132/145	88.66%	399/450
src/itineraries/engine		99.58%	241/242	92.3%	96/104	100%	52/52	99.54%	221/222
src/middleware		88.09%	37/42	84.21%	16/19	100%	8/8	87.8%	36/41
src/nearby		98.56%	137/139	89.28%	50/56	97.61%	41/42	99.16%	119/120
src/posts		99.31%	146/147	95%	38/40	100%	45/45	99.27%	136/137
src/shared		100%	1/1	100%	0/0	100%	0/0	100%	1/1
src/sync		90.14%	64/71	89.28%	25/28	84.21%	16/19	89.85%	62/69

Code coverage generated by istanbul at 2026-10-26T23:42:46.980Z

```
File                     | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------|---------|----------|---------|---------|------------------
All files                |   91.14 |     74.5 |    92.8 |   90.81 |
  app.ts                 |     100 |       50 |     100 |     100 | 38
  activityModel.ts       |     100 |      100 |     100 |     100 |
  activityRoutes.ts      |     100 |      100 |     100 |     100 |
  activityService.ts     |   86.08 |    73.91 |   86.95 |   86.11 | 178,263,269-291
  communityScore.ts      |   93.33 |       50 |     100 |    92.3 | 34
  interests.ts           |     100 |       50 |     100 |     100 | 76-86
  openingHours.ts        |   92.85 |    91.66 |     100 |    92.3 | 31
  placesClient.ts        |   41.17 |        0 |       0 |   41.17 | 49-50,88-141
  reviewModel.ts         |     100 |      100 |     100 |     100 |
  reviewTags.ts          |     100 |    92.85 |     100 |     100 | 38
  authRoutes.ts          |     100 |      100 |     100 |     100 |
  authService.ts         |   94.11 |    92.85 |     100 |   94.11 | 31-36
  googleTokenVerifier.ts |   54.54 |    16.66 |     100 |   54.54 | 17,20-24
  userModel.ts           |     100 |      100 |     100 |     100 |
  env.ts                 |       0 |        0 |       0 |       0 | 1-16
  connection.ts          |     100 |      100 |     100 |     100 |
  disruptionRoutes.ts    |     100 |      100 |     100 |     100 |
  disruptionService.ts   |   83.06 |    56.33 |   88.88 |   83.05 | 79-80,134,151,173,211,249,259-263,274-285
  suggestionModel.ts     |     100 |      100 |     100 |     100 |
  weatherClient.ts       |       0 |        0 |       0 |       0 | 1-70
  friendRoutes.ts        |     100 |      100 |     100 |     100 |
  friendService.ts       |   96.47 |    86.36 |      96 |    96.2 | 51,127-128
  friendshipModel.ts     |     100 |      100 |     100 |     100 |
  groupModel.ts          |     100 |      100 |     100 |     100 |
  groupRoutes.ts         |     100 |      100 |     100 |     100 |
  groupService.ts        |   95.09 |    77.77 |   96.77 |    94.5 | 64,107-112,180
  dayRepair.ts           |   98.57 |    78.81 |     100 |   98.26 | 154,214
  itineraryModel.ts      |     100 |      100 |     100 |     100 |
  itineraryRoutes.ts     |     100 |      100 |     100 |     100 |
  itineraryService.ts    |    93.9 |    81.72 |     100 |   93.06 | 260,295,484,505,512,529,580,614-620
  routesClient.ts        |      44 |        0 |       0 |      44 | 18,36-37,52,57-83
  travelTimes.ts         |      46 |     23.8 |   46.15 |   48.88 | 50,67-69,80-109
  tripCalendar.ts        |     100 |       80 |     100 |     100 | 23
  tripOptions.ts         |     100 |      100 |     100 |     100 |
  visits.ts              |     100 |      100 |     100 |     100 |
  adjust.ts              |   94.73 |      100 |     100 |   94.44 | 18
  candidates.ts          |     100 |    95.83 |     100 |     100 | 26
  daySchedule.ts         |     100 |    92.85 |     100 |     100 | 59
  localSearch.ts         |     100 |    95.23 |     100 |     100 | 18
  planner.ts             |     100 |     86.2 |     100 |     100 | 45,73,139-141
  travelTime.ts          |     100 |       90 |     100 |     100 | 32
  types.ts               |     100 |      100 |     100 |     100 |
  authenticate.ts        |   94.11 |    83.33 |     100 |   94.11 | 28
  errorHandler.ts        |      84 |    84.61 |     100 |   83.33 | 26-27,42-43
  joinRequestModel.ts    |     100 |      100 |     100 |     100 |
  nearbyRoutes.ts        |     100 |      100 |     100 |     100 |
  nearbyService.ts       |   98.05 |    89.28 |   97.05 |   98.82 | 133
  postModel.ts           |     100 |      100 |     100 |     100 |
  postRoutes.ts          |     100 |      100 |     100 |     100 |
  postService.ts         |   98.85 |    94.73 |     100 |   98.73 | 178
  interestTags.ts        |     100 |      100 |     100 |     100 |
  tripSync.ts            |   90.14 |    89.28 |   84.21 |   89.85 | 73,94-96,136,144,150
```

Without mocks, the external service clients are not reached: Google Places (`placesClient.ts`), Google Routes (`routesClient.ts` and the routed part of `travelTimes.ts`), OpenWeather (`weatherClient.ts` and the forecast checks in `disruptionService.ts`), and successful Google token verification. Real calls need API keys and a real Google ID token, so the tests with mocks cover these paths and database failures.

"Uncovered Line #s" refers to lines of the TypeScript source (source maps are on in the back-end `tsconfig.json`).

2.5. Jest Coverage Report Screenshots With and Without Mocks Combined

All files

98.94% Statements `1977/1998` **88.76%** Branches `766/863` **98.8%** Functions `494/500` **98.91%** Lines `1831/1851`

Press *n* or *j* to go to the next uncovered block, *b*, *p* or *k* for the previous block.

Filter:

File		Statements		Branches		Functions		Lines	
src		100%	29/29	100%	2/2	100%	2/2	100%	29/29
src/activities		100%	287/287	87.07%	128/147	100%	54/54	100%	275/275
src/auth		100%	75/75	95.45%	21/22	100%	17/17	100%	74/74
src/config		0%	0/8	0%	0/11	0%	0/2	0%	0/8
src/db		100%	7/7	100%	0/0	100%	3/3	100%	7/7
src/disruptions		100%	171/171	93.5%	72/77	100%	35/35	100%	165/165
src/friends		100%	128/128	97.91%	47/48	100%	36/36	100%	121/121
src/groups		100%	143/143	91.11%	41/45	100%	40/40	100%	132/132
src/itineraries		99.01%	503/508	85.22%	225/264	100%	145/145	98.88%	445/450
src/itineraries/engine		99.58%	241/242	92.3%	96/104	100%	52/52	99.54%	221/222
src/middleware		92.85%	39/42	89.47%	17/19	100%	8/8	92.68%	38/41
src/nearby		99.28%	138/139	91.07%	51/56	97.61%	41/42	100%	120/120
src/posts		100%	147/147	97.5%	39/40	100%	45/45	100%	137/137
src/shared		100%	1/1	100%	0/0	100%	0/0	100%	1/1
src/sync		95.77%	68/71	96.42%	27/28	84.21%	16/19	95.65%	66/69

Code coverage generated by istanbul at 2026-10-26T23:44:34.444Z

```
File                     | % Stmts | % Branch | % Funcs | % Lines | Uncovered Line #s
-------------------------|---------|----------|---------|---------|------------------
All files                |   98.94 |    88.76 |    98.8 |   98.91 |
  app.ts                 |     100 |      100 |     100 |     100 |
  activityModel.ts       |     100 |      100 |     100 |     100 |
  activityRoutes.ts      |     100 |      100 |     100 |     100 |
  activityService.ts     |     100 |     88.4 |     100 |     100 | 52-56,194,211,286
  communityScore.ts      |     100 |       80 |     100 |     100 | 25-26
  interests.ts           |     100 |    66.66 |     100 |     100 | 85-86
  openingHours.ts        |     100 |      100 |     100 |     100 |
  placesClient.ts        |     100 |    76.92 |     100 |     100 | 116-122,141
  reviewModel.ts         |     100 |      100 |     100 |     100 |
  reviewTags.ts          |     100 |    92.85 |     100 |     100 | 38
  authRoutes.ts          |     100 |      100 |     100 |     100 |
  authService.ts         |     100 |    92.85 |     100 |     100 | 48
  googleTokenVerifier.ts |     100 |      100 |     100 |     100 |
  userModel.ts           |     100 |      100 |     100 |     100 |
  env.ts                 |       0 |        0 |       0 |       0 | 1-16
  connection.ts          |     100 |      100 |     100 |     100 |
  disruptionRoutes.ts    |     100 |      100 |     100 |     100 |
  disruptionService.ts   |     100 |    94.36 |     100 |     100 | 98,162-163,277
  suggestionModel.ts     |     100 |      100 |     100 |     100 |
  weatherClient.ts       |     100 |    83.33 |     100 |     100 | 74
  friendRoutes.ts        |     100 |      100 |     100 |     100 |
  friendService.ts       |     100 |    97.72 |     100 |     100 | 169
  friendshipModel.ts     |     100 |      100 |     100 |     100 |
  groupModel.ts          |     100 |      100 |     100 |     100 |
  groupRoutes.ts         |     100 |      100 |     100 |     100 |
  groupService.ts        |     100 |    91.11 |     100 |     100 | 122,219,242-252
  dayRepair.ts           |   98.57 |    78.81 |     100 |   98.26 | 154,214
  itineraryModel.ts      |     100 |      100 |     100 |     100 |
  itineraryRoutes.ts     |     100 |      100 |     100 |     100 |
  itineraryService.ts    |   98.47 |    90.32 |     100 |   98.26 | 295,505,512
  routesClient.ts        |     100 |    92.85 |     100 |     100 | 70
  travelTimes.ts         |     100 |    85.71 |     100 |     100 | 90,105
  tripCalendar.ts        |     100 |       80 |     100 |     100 | 23
  tripOptions.ts         |     100 |      100 |     100 |     100 |
  visits.ts              |     100 |      100 |     100 |     100 |
  adjust.ts              |   94.73 |      100 |     100 |   94.44 | 18
  candidates.ts          |     100 |    95.83 |     100 |     100 | 26
  daySchedule.ts         |     100 |    92.85 |     100 |     100 | 59
  localSearch.ts         |     100 |    95.23 |     100 |     100 | 18
  planner.ts             |     100 |     86.2 |     100 |     100 | 45,73,139-141
  travelTime.ts          |     100 |       90 |     100 |     100 | 32
  types.ts               |     100 |      100 |     100 |     100 |
  authenticate.ts        |   94.11 |    83.33 |     100 |   94.11 | 28
  errorHandler.ts        |      92 |     92.3 |     100 |   91.66 | 26-27
  joinRequestModel.ts    |     100 |      100 |     100 |     100 |
  nearbyRoutes.ts        |     100 |      100 |     100 |     100 |
  nearbyService.ts       |   99.02 |    91.07 |   97.05 |     100 | 71,99,131,135-141
  postModel.ts           |     100 |      100 |     100 |     100 |
  postRoutes.ts          |     100 |      100 |     100 |     100 |
  postService.ts         |     100 |    97.36 |     100 |     100 | 113
  interestTags.ts        |     100 |      100 |     100 |     100 |
  tripSync.ts            |   95.77 |    96.42 |   84.21 |   95.65 | 136,144,150
```

Why the combined coverage is below 100%:

- `src/config/env.ts` **(0%)**: reads the environment variables once at start-up and is imported only by `src/index.ts`, which is excluded from coverage. The tests build the app with explicit configuration (`backend/tests/helpers/testApp.ts`) so they do not depend on a developer's `.env`.
- `src/itineraries/itineraryService.ts` **line 295**: a guard for a planned stop whose activity is not a candidate. The engine schedules only its candidates, so this cannot happen; the guard exists because `Map.get` may return `undefined`.

- `src/itineraries/itineraryService.ts` **lines 505 and 512, and** `src/itineraries/dayRepair.ts` **line 154**: a disruption proposal is dropped if the stop left the day between the timer's check and the proposal (a simultaneous edit), or if the rest of the day no longer fits without it (line 154, surfaced as line 512). Removing a stop from a feasible day keeps it feasible whenever travel times obey the triangle inequality, which our estimates always do; only Google Routes times could break it. Both are race or data guards that cannot be produced reliably through the API.
- `src/itineraries/dayRepair.ts` **line 214**: the tie-break when two ways of making room on a full day drop stops of exactly the same value; the one with less travel wins. A tie needs two places with the same rating, rating count and interests.
- `src/itineraries/engine/adjust.ts` **line 18**: during a re-plan, a reordering ends the day sooner and the freed time goes to another activity. Greedy insertion already places each stop at the least travel, so the short test days never improve by reordering; the engine's unit tests cover the reorder step (`backend/tests/unit/localSearch.test.ts` and `dayRepair.test.ts`).
- `src/sync/tripSync.ts` **lines 136, 144 and 150**: `.catch` handlers on Socket.io events. Subscribing and sharing catch their own errors (tested with mocks), and leaving a room cannot fail with the in-memory adapter, so these only log an unexpected bug instead of letting an unhandled rejection stop the server.
- `src/nearby/nearbyService.ts` **(99% of statements)**: the constructor's default clock. The tests inject their own clock (`backend/tests/helpers/testApp.ts`) so dates are predictable.
- `src/middleware/errorHandler.ts` **lines 26-27**: passes an error to Express's default handler if a response was partly sent. No route writes before failing, so this is unreachable through the API; it follows the Express documentation's error-handling pattern.
- `src/middleware/authenticate.ts` **line 28**: rejects a protected route reached without the session middleware. Every protected router installs `requireSession`, so this is a defensive check.
- **Remaining branch gaps** (for example `interests.ts` 85-86, `placesClient.ts` 116-122, `weatherClient.ts` 74, `tripCalendar.ts` 23, the engine files): destructuring defaults and `??` fallbacks for fields Google Places, Google Routes or OpenWeather may omit, required by TypeScript's strict null checks.

---

3. Tests of Non-Functional Requirements

3.1. Test Locations in Git

Non-Functional Requirement	Location in Git
NFR 1: Itinerary generation time	`backend/tests/nfr/itineraryGenerationTime.test.ts`
NFR 2: Live update latency	`backend/tests/nfr/liveUpdateLatency.test.ts`
NFR 3: Bounded Google Places usage	`backend/tests/nfr/placesUsage.test.ts`

3.2. Test Verification and Logs

- **NFR 1: Itinerary generation time**
  - **Verification:** The test loads the 30-activity Vancouver catalogue plus 30 synthetic downtown activities, so pruning keeps the maximum of 40 candidates. It sends 20 consecutive `POST /api/itineraries/generate` requests for a 5-day trip (08:00 to 21:00, packed pace, all interests) through the real Express app and an in-memory MongoDB, times each from send to response, and asserts that the 95th percentile is under 3,000 ms. It also checks that every response is 201 with 5 days and that the engine saw 40 candidates.
  - **Log Output**

```
==> Running backend NFR tests (tests/nfr/) ...
  console.info
    NFR itinerary generation: 20 runs, 5 days, 40 candidates, 33 stops
    p50 345 ms, p95 416 ms, max 457 ms (limit 3000 ms)

Test Suites: 1 passed, 1 total
Tests:       1 passed, 1 total
==> Backend NFR tests finished.
```

- **NFR 2: Live update latency**

- **Verification:** The test creates a 4-member group with a 3-day packed Vancouver trip, serves the app and its Socket.io hub on a local port, and subscribes the other three members. The owner sends 20 adjustments, alternating the cheapest (lock or unlock a stop) and the most expensive (re-plan a day). For each, it times every member from the request to receiving `itineraryUpdated`, which includes the re-plan, the database write and the push. All 60 deliveries must take under 2,000 ms.
- **Log Output**

```
==> Running backend NFR tests (tests/nfr/) ...
  console.info
    NFR live update latency: 20 adjustments x 3 other members = 60 deliveries
    p50 53 ms, p95 77 ms, max 98 ms (limit 2000 ms)
```

- **NFR 3: Bounded Google Places usage**
  - **Verification:** The test runs the real Express app and an in-memory MongoDB with a fake Google Places client that counts every call, and a clock the test moves forward. For 6 days, two travellers search Vancouver four times a day and generate a 2-day itinerary once a day. The clock then passes 7 days for one more search, and a final itinerary is generated with a lodging. The test asserts 7 text searches (one per interest tag) for the first search, none over the 6 days, 7 for the refresh after 7 days, and exactly 1 for the lodging, and that every request succeeds.
  - **Log Output**

```
==> Running backend NFR tests (tests/nfr/) ...
  console.info
    NFR Places usage: 7 calls for the first fetch; 0 more over 6 days of 24 searches and 6 generations; 7 for the
refresh after 7 days; 1 for a lodging lookup

Test Suites: 3 passed, 3 total
Tests:       3 passed, 3 total
==> Backend NFR tests finished.
```

---

4. Front-end Test Specification

4.1. Location in Git of Front-end Test Suite:

`frontend/app/src/androidTest/java/com/example/cpen321application/e2e/`

There is one test class per use case. Shared steps (waiting for the back-end, picking dates and times, scrolling) are in `androidTest/.../testing/E2eTest.kt`, outside the `e2e` folder so the course script does not run it as a test.

The tests find elements by stable test tags (`ui/common/TestTags.kt`), so they depend on wording only where they check it. As the course requires, each test starts on the home screen signed in and runs against the real back-end without mocks. Each test creates the trips it needs, and dates are computed from the day it runs.

How to run them: start the back-end and the emulator, run `./scripts/run-frontend-e2e-tests.sh` (or `.\scripts\run-frontend-e2e-tests.ps1` on Windows) from the repository root, sign in when prompted, and press Enter. The script runs `:app:connectedDebugAndroidTest` for every class in `e2e`. To run one test, use Android Studio or, with both APKs installed, `adb shell am instrument -w -e class com.example.cpen321application.e2e.<TestClass> com.example.cpen321application.test/androidx.test.runner.AndroidJUnitRunner`, which captured the logs below (Pixel 9 emulator, Android 16).

4.2. Tests

- **Use Case: Generate Itinerary** (formal specification): `GenerateItineraryTest.kt`
  - **Expected Behaviors:**

Scenario Steps	Test Case Steps
1. The organizer taps "Plan a trip" on the home screen.	Click the node tagged `homePlanTrip`.

Scenario Steps	Test Case Steps
2. The app shows the Trip Builder, with the destination and interests prefilled from the organizer's profile.	Check the node tagged `tripCityField` is displayed. Check the button tagged `generateButton` with text "Generate itinerary" is displayed.
3a. The destination is empty.	Clear `tripCityField`. Click `generateButton`.
3a1. The app shows "Enter a destination city." and does not generate.	Check `tripError` has text "Enter a destination city.". Check the Trip Builder is still displayed.
3b. No dates are chosen.	Type "Vancouver" in `tripCityField`. Click `generateButton`.
3b1. The app shows "Choose your trip dates." and does not generate.	Check `tripError` has text "Choose your trip dates.".
3c. The trip is longer than 7 days.	Click `tripDatesButton`. Choose a start date and an end date 8 days later. Click "Done". Click `generateButton`.
3c1. The app shows "Trips can be at most 7 days." and does not generate.	Check `tripError` has text "Trips can be at most 7 days.".
3. The organizer enters the destination city and chooses the trip's first and last day.	Click `tripDatesButton`. Choose a start date 7 days from today and an end date 9 days from today. Click "Done". Check `tripDatesButton` shows "3 days".
4a. The day ends less than 2 hours after it starts.	Open the "Start" dropdown and choose 17:00. Click `generateButton`.
4a1. The app shows "Your day must end at least 2 hours after it starts." and does not generate.	Check `tripError` has text "Your day must end at least 2 hours after it starts.". Open the "Start" dropdown and choose 09:00.
4. The organizer optionally changes the daily times, arrival and departure, lodging, interests and pace.	Open the "Arrive" dropdown and choose 13:00. Click the "Museums" chip. Click the "Relaxed" segment.
6. The organizer taps "Generate itinerary".	Click `generateButton`.
7. The system ranks and schedules the city's activities into each day.	Wait until a node tagged `itineraryDay` is displayed (up to 10 s).
8. The app shows a day-by-day timeline.	Wait until a stop's "HH:MM to HH:MM" line (inside a node tagged `itineraryStop`) is displayed. Check the first day's first stop starts at or after 13:00. Scroll the timeline to the text "Back at lodging by", then to the headings "Day 1 of 3", "Day 2 of 3" and "Day 3 of 3".
7b. The lodging cannot be found.	Go back to home, click `homePlanTrip`, replace the text in `tripCityField` with "Vancouver", choose the same dates, type "no such place zzzz" in `tripLodgingField`, click `generateButton`.
7b1-7b2. The system plans from the city centre and the itinerary shows a warning.	Wait until a node tagged `itineraryWarning` is displayed. Check the text inside it contains "Could not find". Check the text "From Vancouver centre" is displayed.

Failure scenarios 7a (no activities for the city) and 7f (TimeTravel unreachable) need a city without activities or a stopped back-end, so they are verified manually; the front-end tests run against the real deployed back-end without mocks.

- **Test Logs:**

```
$ adb shell am instrument -w -e class com.example.cpen321application.e2e.GenerateItineraryTest
com.example.cpen321application.test/androidx.test.runner.AndroidJUnitRunner

com.example.cpen321application.e2e.GenerateItineraryTest:.

Time: 16.875

OK (1 test)
```

- **Use Case: Rate & Review Activity** (formal specification): `RateReviewActivityTest.kt`
  - **Expected Behaviors:**

Scenario Steps	Test Case Steps
1. On an activity's card in Explore & review, the traveller taps Rate.	Click the tab tagged `tabExplore`. Replace the text in `exploreCityField` with "Vancouver" and click `exploreSearchButton`. Wait until a node tagged `activityCard` is displayed (up to 10 s) and read the first card's activity name. Click the first `rateButton`.
2. The app opens a rating dialog with five stars, a review field and optional tags. Submit is disabled.	Check the text "Rate <activity>" is displayed. Check `submitRatingButton` is not enabled.
4b. The traveller taps Cancel.	Click "Cancel".
4b1. The dialog closes and nothing is saved.	Wait until `submitRatingButton` no longer exists. Click the first `rateButton` again.
3. The traveller selects a star rating. Submit becomes enabled.	Click `ratingStar4`. Check `submitRatingButton` is enabled.
4a. The traveller types more than 500 characters.	Type 520 characters into `reviewField`.
4a1. The review field stops accepting characters at 500.	Check `reviewField` holds exactly 500 characters.
4. The traveller optionally writes a review and selects tags, then taps Submit.	Replace the text in `reviewField` with "Great views, worth the walk.". Click the "Morning" and "Outdoor" chips. Click `submitRatingButton`.
5. The system saves the review and recomputes the activity's score and tags.	Wait until the text "Thanks! Your review of <activity> was saved." is displayed (up to 25 s, so the app can read the device's location first).
6. The app closes the dialog, updates the card and shows the confirmation.	Wait until `submitRatingButton` no longer exists. Click the `reviewsButton` on the card of <activity>. Check "Reviews of <activity>" and "Great views, worth the walk." are displayed. Click "Close".

    Failure scenarios 5a (the activity no longer exists) and 5b (TimeTravel cannot be reached) need the activity deleted mid-dialog or the back-end stopped, which tests against the real back-end cannot cause. The back-end interface tests cover them (404 for a missing activity, and database failures with mocks).

  - **Test Logs:**

```
$ adb shell am instrument -w -e class com.example.cpen321application.e2e.RateReviewActivityTest
com.example.cpen321application.test/androidx.test.runner.AndroidJUnitRunner

com.example.cpen321application.e2e.RateReviewActivityTest:.

Time: 13.368

OK (1 test)
```

- **Use Case: View & Adjust Itinerary** (formal specification): `ViewAdjustItineraryTest.kt`
  - **Expected Behaviors:**

Scenario Steps	Test Case Steps
(Set-up: a trip to adjust.)	Click `homePlanTrip`. Replace the text in `tripCityField` with "Vancouver". Choose a start date 10 days from today and an end date a day later. Open the "Start" dropdown and choose 11:00. Click `generateButton`, wait until a node tagged `itineraryDay` is displayed, and go back to home.
1. The traveller opens a trip from the home screen's recent trips.	Click the first node tagged `recentTrip` (the newest trip).

Scenario Steps	Test Case Steps
2. The app shows the itinerary timeline, connects to live updates and shows "Live".	Wait until `liveStatus` contains "LIVE:" and a stop's "HH:MM to HH:MM" line is displayed. Read the first two stops' names (from their menu buttons, "Options for <stop>") and the first stop's times.
3. The traveller opens a stop's menu and locks it.	Click the first `stopActionsButton`, then `stopActionLock`.
4-5. The system re-plans; locked stops keep their times, and the traveller's app shows the change.	Wait until a node tagged `stopLocked` is displayed. Check the first stop's times are unchanged.
4a. No feasible plan keeps the requested change.	Click the second `stopActionsButton`, then `stopActionMoveEarlier`. (The second stop would have to fit before the locked first stop, which starts as soon as the day does.)
4a1. The app explains why and keeps the previous plan.	Wait until the text "<second stop> cannot go there" is displayed. Check the first two stops are still <first stop> and <second stop>.
3. The traveller removes a stop.	Click the second `stopActionsButton`, then `stopActionRemove`.
4-5. The removed stop's time goes to another activity; locked stops keep their times.	Wait until no button "Options for <second stop>" exists. Check the first stop is still <first stop>, with the same times.
3. The traveller unlocks a stop.	Click the first `stopActionsButton`, then `stopActionLock` (now "Unlock").
4-5. The stop is no longer locked.	Wait until no node tagged `stopLocked` exists.

Step 5 for the other members needs a second device; the Socket.io interface tests and the NFR 2 test cover it. Failure scenarios 2a (missing itinerary or not a member), 2b (lost connection) and 4b (another member changed the trip first) need a second account, a lost network or a concurrent edit; the back-end interface tests cover their server side (404, the live connection, and 409 for an out-of-date version).

- **Test Logs:**

```
$ adb shell am instrument -w -e class com.example.cpen321application.e2e.ViewAdjustItineraryTest
com.example.cpen321application.test/androidx.test.runner.AndroidJUnitRunner

com.example.cpen321application.e2e.ViewAdjustItineraryTest:.

Time: 10.745

OK (1 test)
```